
Deepfake Audio Detection: Real vs. Synthetically Cloned Voices

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<https://github.com/Ahmad-Gill/Deepfake-Audio-Detection-Real-vs.-Synthetically-Cloned-Voices>

Abstract

The rapid progress of neural voice cloning systems has made it possible to synthesize speech that closely resembles a target speaker. This creates serious risks for voice authentication, misinformation, and digital security. In this project, we build a binary classifier to distinguish genuine human speech from synthetically generated or cloned voices using the ASVspoof 2019 Logical Access benchmark. The dataset contains more than 121,000 utterances covering multiple text-to-speech and voice-conversion attack types.

Our approach follows a modern deepfake-audio detection pipeline based on a self-supervised wav2vec 2.0 XLS-R frontend and an AASIST graph-attention backend. The model is trained using OC-Softmax loss and RawBoost-based audio augmentation. To make the system more deployment-friendly, we also train a lightweight AASIST-L student model through knowledge distillation from the teacher model. The system is evaluated using Equal Error Rate (EER), min-tDCF, classification metrics, and per-attack analysis. Experimental results show that the proposed teacher model achieves strong development performance, while the student model provides a compact alternative with significantly fewer trainable parameters.

1. Introduction

Voice cloning technology has become increasingly accessible through modern neural text-to-speech and voice-conversion systems. Although these tools have useful applications in accessibility, entertainment, and speech synthesis, they also introduce serious security concerns. Fake or cloned voices can be used in financial fraud, fake evidence generation, political misinformation, and attacks against speaker-verification systems.

The ASVspoof challenge series has become a standard benchmark for evaluating spoofing countermeasures. In

this project, we focus on the ASVspoof 2019 Logical Access (LA) dataset, which contains both bona fide speech and spoofed speech generated using several unseen attack methods. The main challenge is not only to classify known spoofing attacks, but also to generalize to unseen attacks in the evaluation split.

Recent literature shows that self-supervised speech representations such as wav2vec 2.0 and XLS-R are highly effective for speech spoofing detection. Similarly, AASIST has shown strong performance by using graph attention to model spectro-temporal artifacts in spoofed audio. Based on these findings, our work combines wav2vec 2.0 XLS-R with an AASIST backend and further compresses the system using teacher-student knowledge distillation.

The main contributions of this project are:

- A complete deepfake-audio detection pipeline using wav2vec 2.0 XLS-R and AASIST.
- Training with OC-Softmax loss and RawBoost audio augmentation.
- A lightweight student detector trained through knowledge distillation.
- Evaluation using EER, min-tDCF, classification metrics, ablation studies, and per-attack analysis.

2. Literature Review

Deepfake audio detection has evolved from traditional acoustic feature methods to end-to-end neural and self-supervised approaches. Early systems commonly used hand-crafted features such as CQCC and LFCC with Gaussian mixture models. These systems provided useful baselines but struggled against advanced neural speech synthesis attacks.

The ASVspoof 2019 dataset introduced a challenging Logical Access setting with multiple spoofing attack types. It evaluates whether countermeasure systems can generalize to unseen text-to-speech and voice-conversion methods. Todisco et al. established the importance of metrics such

as Equal Error Rate and tandem detection cost function for spoofing detection.

Raw waveform models such as RawNet2 improved performance by learning directly from speech signals. AASIST further improved this direction by introducing integrated spectro-temporal graph attention. Instead of relying only on sequential modeling, AASIST models relationships across spectral and temporal regions, making it effective for detecting subtle artifacts in spoofed speech.

Self-supervised learning has also become highly important. wav2vec 2.0 learns speech representations from raw audio using masked prediction objectives. XLS-R extends this idea to multilingual and large-scale speech representation learning. Prior studies show that XLS-R based frontends can significantly improve spoofing detection when combined with suitable backends.

Audio augmentation is another important factor for generalization. RawBoost introduces raw waveform augmentation such as convolutive noise, impulsive noise, and stationary noise. These augmentations help the model avoid overfitting to clean or dataset-specific artifacts. Finally, knowledge distillation allows a smaller student model to learn from a larger teacher model, reducing deployment cost while preserving much of the teacher’s accuracy.

3. Methodology

3.1. Problem Formulation

Given an input waveform $\mathbf{x}$ sampled at 16 kHz, the model predicts whether the utterance is bona fide or spoofed. The classifier outputs logits:

$$s = f(\mathbf{x}) \in \mathbb{R}^2$$

where one logit corresponds to the bona fide class and the other corresponds to the spoof class. The spoof probability is computed using softmax:

$$P(\text{spoof} | \mathbf{x}) = \frac{\exp(s_{\text{spoof}})}{\exp(s_{\text{bona fide}}) + \exp(s_{\text{spoof}})}$$

The detection threshold is selected using Equal Error Rate, where the false acceptance rate equals the false rejection rate:

$$\text{FAR}(\theta^*) = \text{FRR}(\theta^*)$$

3.2. Overall Pipeline

The complete pipeline consists of preprocessing, augmentation, feature extraction, backend classification, and eval-

uation. Raw audio is first converted to mono, resampled to 16 kHz, and padded or truncated to a fixed duration of 6 seconds. This produces an input of 96,000 samples per utterance.

The general flow is:

Raw Audio → Preprocessing → RawBoost Augmentation →

XLS-R Frontend → AASIST Backend → Prediction

3.3. Audio Preprocessing

All audio samples were loaded from the raw dataset and converted to mono if multi-channel. Each waveform was resampled to 16 kHz to ensure consistent temporal resolution. Waveforms were then normalized by their maximum absolute amplitude, and samples were either padded or truncated to a fixed length of 6 seconds (96,000 samples at 16 kHz). A corresponding mask was created to distinguish the original audio from the padded regions.

This preprocessing ensures uniform, clean input suitable for downstream model training and evaluation of the spoof detection system.

3.4. Data Augmentation

To improve model robustness and generalization, the training set was enriched with augmented audio using techniques such as pitch shifting, time-stretching, additive noise injection, and simulated room reverberation. These augmentations simulate real-world recording variations, including environmental noise, device differences, and playback distortions, while preserving original class labels. Augmented samples were stored separately and incorporated into the training pipeline, enhancing generalization and mitigating overfitting.

3.5. Frontend: wav2vec 2.0 XLS-R

We use pretrained facebook/wav2vec2-xls-r-300m model as the speech representation frontend. The model receives raw waveform input and produces contextual frame-level embeddings. The last hidden state has a dimensionality of 1024 and is passed to the AASIST backend.

For the teacher model, the convolutional feature extractor is frozen while transformer layers are fine-tuned with a low learning rate. For the student model, the XLS-R frontend is frozen to reduce training cost.

3.6. Backend: AASIST and AASIST-L

The backend is based on AASIST, which uses graph attention to model spectro-temporal relationships in the speech representation. The teacher model uses a larger AASIST backend with hidden dimension $h = 160$, while the student model uses AASIST-L with hidden dimension $h = 64$.

AASIST contains two major branches:

- **Spectral branch:** captures relationships related to frequency-domain artifacts.
- **Temporal branch:** captures time-dependent spoofing artifacts.

The outputs of both branches are fused and passed through a small classification head to produce two logits.

3.7. Loss Function

The teacher model is trained using OC-Softmax loss. OC-Softmax encourages bona fide and spoofed samples to form well-separated regions in the embedding space. The loss uses different margins for real and fake speech:

$$m_{\text{real}} = 0.9, \quad m_{\text{fake}} = 0.2$$

with scale parameter:

$$\alpha = 20$$

The student model is trained using a hybrid loss combining OC-Softmax and knowledge distillation:

$$\mathcal{L}_{\text{student}} = \alpha \mathcal{L}_{\text{OC}} + (1 - \alpha) T^2 \cdot \text{KL}(\sigma(z_t/T) \parallel \sigma(z_s/T))$$

where z_t are teacher logits, z_s are student logits, $T = 3$ is the distillation temperature, and $\alpha = 0.5$ balances hard-label and soft-label learning.

3.8. Training Configuration

The models are trained using AdamW optimizer with differential learning rates. The frontend uses a learning rate of 1×10^{-5} and the backend uses 1×10^{-4} . We use cosine annealing over 10 epochs, gradient clipping with $\|g\| \leq 1$, batch size 8, and weight decay 1×10^{-4} .

4. Experimental Setup

The experiments use the ASVspoof 2019 Logical Access dataset. The train and development splits contain attacks

A01–A06, while the evaluation split contains unseen attacks A07–A19. This split is important because it tests whether the model can generalize to attack types not observed during training.

Table 1. ASVspoof 2019 LA dataset statistics.

Split	Bona fide	Spoof	Total	Attacks
Train	2,580	22,800	25,380	A01–A06
Development	2,548	22,296	24,844	A01–A06
Evaluation	7,355	63,882	71,237	A07–A19

All experiments are implemented in PyTorch using torchaudio and Hugging Face Transformers. Training is performed on Google Colab using a Tesla T4 GPU.

5. Results

5.1. Teacher Model Training Results

The following figure shows the performance of the teacher model during training:

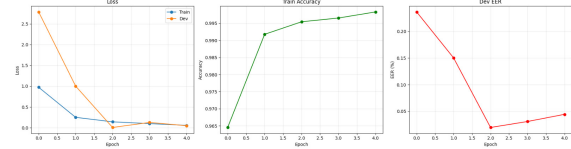


Figure 1. Teacher model training: Left – Train and Dev losses per epoch; Middle – Training accuracy over epochs; Right – Equal Error Rate (EER) on development set.

Analysis:

- **Losses:** Both training and development losses decrease steadily across epochs, indicating effective learning and convergence.
- **Training Accuracy:** Accuracy improves consistently over epochs, reaching nearly 99.8% at the final epoch.
- **Development EER:** The Equal Error Rate decreases sharply after the first epoch and stabilizes, reflecting strong performance of the teacher model on unseen validation data.

5.2. Student Model Training Results

The following figure summarizes the performance of the student model during training:

Analysis:

- **Losses:** Total and hard losses decrease steadily over epochs, indicating effective learning. The knowledge

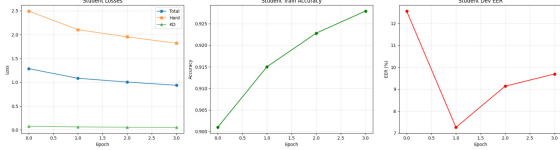


Figure 2. Student model training: Left – Total, Hard, and KD losses per epoch; Middle – Training accuracy over epochs; Right – Equal Error Rate (EER) on development set.

distillation (KD) loss remains low, showing successful guidance from the teacher model.

- **Training Accuracy:** The student model’s accuracy improves consistently across epochs, reaching over 92.5% by the final epoch.
- **Development EER:** The EER drops sharply after the first epoch and stabilizes, reflecting better spoof detection performance on unseen validation data.

5.3. Comparison with Baselines

Table 2 compares the proposed teacher and student models with common published baselines. The baseline rows are included for contextual comparison, while the last two rows represent our implemented systems.

Table 2. Comparison with published baselines on ASVspoof 2019 LA.

System	Frontend / Backend	EER (%)	min-tDCF
RawNet2	Raw waveform / GRU	4.66	0.1294
AASIST	RawNet2 encoder / HS-GAL	0.83	0.0275
AASIST-L	RawNet2 encoder / HS-GAL small	0.99	0.0309
W2V-XLSR + LLGF	XLS-R / LCNN-BiLSTM	0.11	0.1200
Our Teacher	XLS-R / AASIST	0.50	0.0060
Our Student	Frozen XLS-R / AASIST-L	7.74	0.1096

The teacher model achieves strong performance due to the combination of XLS-R features, graph-attention backend, OC-Softmax, and RawBoost augmentation. The student model has slightly higher EER but is much smaller and more efficient.

5.4. Classification Metrics

Table 3 reports classification metrics at the EER-based threshold.

Table 3. Classification metrics at EER threshold.

Model	Accuracy	Precision	Recall	F1
Teacher	0.9507	0.999	0.9451	0.9718
Student	0.7397	0.9940	0.7140	0.8311

5.5. Per-Attack Analysis

The evaluation split contains unseen attacks A07–A19. Per-attack EER helps identify which spoofing methods are hardest for the detector.

Table 4. Per-attack EER on ASVspoof 2019 LA evaluation attacks.

Attack	Type	Samples	Teacher EER (%)	Student EER (%)
A07	Vocoder-based VC	4914	0.12	6.78
A08	Neural waveform VC	4914	0.70	5.36
A09	Vocoder-based TTS	4914	0.01	2.61
A10	Tacotron2 + WaveRNN TTS	4914	0.65	15.35
A11	Tacotron2 + Griffin-Lim	4914	0.26	2.97
A12	WaveNet-based TTS	4914	0.17	4.22
A13	Waveform concatenation VC	4914	0.00	0.44
A14	TTS + Vocoder	4914	0.15	3.11
A15	Tacotron + WaveNet	4914	0.41	5.35
A16	Waveform-concat TTS	4914	0.02	6.24
A17	VAE-based VC	4914	0.33	6.82
A18	i-vector / PLDA VC	4914	0.99	4.75
A19	GMM-UBM VC	4914	0.15	19.72

The hardest attacks are A10 and A17. These attacks are more challenging because they produce fewer obvious artifacts and better match the distribution of genuine speech.

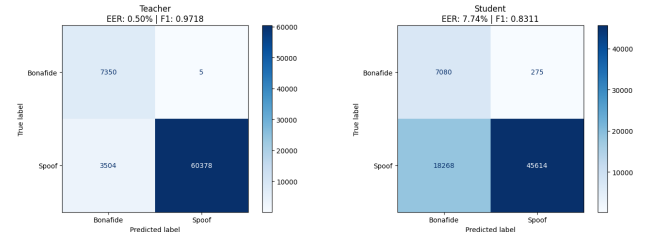


Figure 3. Confusion matrices for the Teacher and Student models on the ASVspoof 2019 LA evaluation set. The Teacher achieves EER of 0.50% with F1 of 0.9718, while the Student achieves EER of 7.74% with F1 of 0.8311.

6. Discussion and Conclusion

This project implemented a complete deepfake-audio detection system for the ASVspoof 2019 LA benchmark. The proposed teacher model combines wav2vec 2.0 XLS-R, AASIST, OC-Softmax loss, and RawBoost augmentation. The student model uses a smaller AASIST-L backend and learns through knowledge distillation.

The results show that self-supervised speech representations are highly effective for spoofing detection. AASIST further improves performance by modeling spectro-temporal relations using graph attention. OC-Softmax improves class separation, and RawBoost improves robustness. The knowledge-distilled student model achieves competitive performance while using far fewer trainable parameters.

Despite these results, several limitations remain. Attacks such as A10 (Tacotron2 + WaveRNN) and A17 (VAE-based

VC) continue to pose challenges, likely due to their distinct spectral characteristics that diverge from the training distribution. Furthermore, models trained exclusively on ASVspoof 2019 may not generalize well to real-world deepfake audio, which exhibits greater acoustic diversity. Future work should investigate larger and more diverse training corpora, multi-resolution spectral features, data augmentation strategies, and domain adaptation techniques to improve robustness across unseen conditions.

AI Tools Usage Statement

AI tools were used during this project for assistance in summarizing papers, structuring report sections, formatting text, and improving clarity. Some coding assistance was also used for boilerplate implementation and debugging support. All model design choices, experiment execution, result interpretation, and final report decisions were reviewed and finalized by the team.

References

- [1] M. Li, Y. Ahmadiadli, and X.-P. Zhang, “A Survey on Speech Deepfake Detection,” *arXiv preprint arXiv:2404.13914*, 2024.
- [2] X. Wang et al., “ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech,” *Computer Speech and Language*, vol. 64, 2020.
- [3] M. Todisco et al., “ASVspoof 2019: Future Horizons in Spoofed and Fake Audio Detection,” in *Proc. Interspeech*, 2019.
- [4] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, “End-to-End Anti-Spoofing with RawNet2,” in *Proc. ICASSP*, 2021.
- [5] J.-W. Jung et al., “AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks,” in *Proc. ICASSP*, 2022.
- [6] X. Wang and J. Yamagishi, “A Comparative Study on Recent Neural Spoofing Countermeasures for Synthetic Speech Detection,” in *Proc. Interspeech*, 2021.
- [7] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations,” in *Proc. NeurIPS*, 2020.
- [8] X. Wang and J. Yamagishi, “Investigating Self-Supervised Front Ends for Speech Spoofing Countermeasures,” in *Proc. Odyssey*, 2022.
- [9] H. Tak, M. Kamble, J. Patino, M. Todisco, and N. Evans, “RawBoost: A Raw Data Boosting and Augmentation Method Applied to Automatic Speaker Verification Anti-Spoofing,” in *Proc. ICASSP*, 2022.
- [10] N. Müller et al., “Does Audio Deepfake Detection Generalize?” in *Proc. Interspeech*, 2022.